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normal instructions



more bytes



undocumented



Z180 only

Main Instructions

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	nop	ld bc,n <i>n</i>	ld (b c),a	inc bc	inc b	dec b	ld b,n	rlca	ex af,a f'	add hl, bc	ld a,(b c)	dec bc	inc c	dec c	ld c,n	rrca
1	djnz <i>d</i>	ld de,n <i>n</i>	ld (d e),a	inc de	inc d	dec d	ld d,n	rla	jr <i>d</i>	add hl, de	ld a,(d e)	dec de	inc e	dec e	ld e,n	rra
2	jr nz, <i>d</i>	ld hl,n <i>n</i>	ld (n),hl	inc hl	inc h	dec h	ld h,n	daa	jr z, <i>d</i>	add hl, hl	ld hl, (nn)	dec hl	inc l	dec l	ld l,n	cpl
3	jr nc, <i>d</i>	ld sp,n <i>n</i>	ld (n),a	inc sp	inc (h l)	dec (h l)	ld (h l),n	scf	jr c, <i>d</i>	add hl, sp	ld a,(n)	dec sp	inc a	dec a	ld a,n	ccf
4	ld b,b	ld b,c	ld b,d	ld b,e	ld b,h	ld b,l	ld b,(h l)	ld b,a	ld c,b	ld c,c	ld c,d	ld c,e	ld c,h	ld c,l	ld c,(h l)	ld c,a
5	ld d,b	ld d,c	ld d,d	ld d,e	ld d,h	ld d,l	ld d,(h l)	ld d,a	ld e,b	ld e,c	ld e,d	ld e,e	ld e,h	ld e,l	ld e,(h l)	ld e,a

6	ld h,b	ld h,c	ld h,d	ld h,e	ld h,h	ld h,l	ld h,(h l)	ld h,a	ld l,b	ld l,c	ld l,d	ld l,e	ld l,h	ld l,l	ld l,(h l)	ld l,a
7	ld (h l),b	ld (h l),c	ld (h l),d	ld (h l),e	ld (h l),h	ld (h l),l	halt	ld (h l),a	ld a,b	ld a,c	ld a,d	ld a,e	ld a,h	ld a,l	ld a,(h l)	ld a,a
8	add a,b	add a,c	add a,d	add a,e	add a,h	add a,l	add a, (hl)	add a,a	adc a,b	adc a,c	adc a,d	adc a,e	adc a,h	adc a,l	adc a, (hl)	adc a,a
9	sub b	sub c	sub d	sub e	sub h	sub l	sub (h l)	sub a	sbc a,b	sbc a,c	sbc a,d	sbc a,e	sbc a,h	sbc a,l	sbc a, (hl)	sbc a,a
A	and b	and c	and d	and e	and h	and l	and (h l)	and a	xor b	xor c	xor d	xor e	xor h	xor l	xor (h l)	xor a
B	or b	or c	or d	or e	or h	or l	or (hl)	or a	cp b	cp c	cp d	cp e	cp h	cp l	cp (hl)	cp a
C	ret nz	pop bc	jp nz, <i>n</i> <i>n</i>	jp <i>nn</i>	call <i>n</i> z, <i>nn</i>	push bc	add a, <i>n</i>	rst 00h	ret z	ret	jp z, <i>nn</i>	Bit	call z, <i>nn</i>	call <i>nn</i>	adc a, <i>n</i>	rst 08h
D	ret nc	pop de	jp nc, <i>n</i> <i>n</i>	out (<i>n</i>),a	call <i>n</i> c, <i>nn</i>	push de	sub <i>n</i>	rst 10h	ret c	exx	jp c, <i>nn</i>	in a, (<i>n</i>)	call c, <i>nn</i>	IX	sbc a, <i>n</i>	rst 18h
E	ret po	pop hl	jp po, <i>n</i> <i>n</i>	ex (s p),hl	call p o, <i>nn</i>	push hl	and <i>n</i>	rst 20h	ret pe	jp (hl)	jp pe, <i>n</i> <i>n</i>	ex de,h l	call p e, <i>nn</i>	Misc.	xor <i>n</i>	rst 28h
F	ret p	pop af	jp p, <i>nn</i>	di	call p, <i>nn</i>	push af	or <i>n</i>	rst 30h	ret m	ld sp,h l	jp m, <i>nn</i>	ei	call m, <i>nn</i>	IY	cp <i>n</i>	rst 38h

Misc. Instructions (ED)

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	in0 b, (<i>n</i>)	out0 (<i>n</i>),b			tst b				in0 c, (<i>n</i>)	out0 (<i>n</i>),c			tst c			
1	in0 d, (<i>n</i>)	out0 (<i>n</i>),d			tst d				in0 e, (<i>n</i>)	out0 (<i>n</i>),e			tst e			
2	in0 h, (<i>n</i>)	out0 (<i>n</i>),h			tst h				in0 l, (<i>n</i>)	out0 (<i>n</i>),l			tst l			

3					tst (h l)				in0 a, (n)	out0 (n),a			tst a			
4	in b, (c)	out (c),b	sbc hl, bc	ld (n n),bc	neg	retn	im 0	ld i,a	in c, (c)	out (c),c	adc hl, bc	ld bc, (nn)	mlt bc	reti		ld r,a
5	in d, (c)	out (c),d	sbc hl, de	ld (n n),de			im 1	ld a,i	in e, (c)	out (c),e	adc hl, de	ld de, (nn)	mlt de		im 2	ld a,r
6	in h, (c)	out (c),h	sbc hl, hl	ld (n n),hl	tst n			rrd	in l, (c)	out (c),l	adc hl, hl	ld hl, (nn)	mlt hl			rld
7	in (c)	out (c),0	sbc hl, sp	ld (n n),sp	tstio n		slp		in a, (c)	out (c),a	adc hl, sp	ld sp, (nn)	mlt sp			
8				otim								otdm				
9				otimr								otdmr				
A	ldi	cpi	ini	outi					ldd	cpd	ind	outd				
B	ldir	cpir	inir	otir					lddr	cpdr	indr	otdr				

Bit Instructions (CB)

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	rlc b	rlc c	rlc d	rlc e	rlc h	rlc l	rlc (h l)	rlc a	rrc b	rrc c	rrc d	rrc e	rrc h	rrc l	rrc (h l)	rrc a
1	rl b	rl c	rl d	rl e	rl h	rl l	rl (hl)	rl a	rr b	rr c	rr d	rr e	rr h	rr l	rr (hl)	rr a
2	sla b	sla c	sla d	sla e	sla h	sla l	sla (h l)	sla a	sra b	sra c	sra d	sra e	sra h	sra l	sra (h l)	sra a
3	sll b	sll c	sll d	sll e	sll h	sll l	sll (h l)	sll a	srl b	srl c	srl d	srl e	srl h	srl l	srl (h l)	srl a

4	bit 0,b	bit 0,c	bit 0,d	bit 0,e	bit 0,h	bit 0,l	bit 0, (hl)	bit 0,a	bit 1,b	bit 1,c	bit 1,d	bit 1,e	bit 1,h	bit 1,l	bit 1, (hl)	bit 1,a
5	bit 2,b	bit 2,c	bit 2,d	bit 2,e	bit 2,h	bit 2,l	bit 2, (hl)	bit 2,a	bit 3,b	bit 3,c	bit 3,d	bit 3,e	bit 3,h	bit 3,l	bit 3, (hl)	bit 3,a
6	bit 4,b	bit 4,c	bit 4,d	bit 4,e	bit 4,h	bit 4,l	bit 4, (hl)	bit 4,a	bit 5,b	bit 5,c	bit 5,d	bit 5,e	bit 5,h	bit 5,l	bit 5, (hl)	bit 5,a
7	bit 6,b	bit 6,c	bit 6,d	bit 6,e	bit 6,h	bit 6,l	bit 6, (hl)	bit 6,a	bit 7,b	bit 7,c	bit 7,d	bit 7,e	bit 7,h	bit 7,l	bit 7, (hl)	bit 7,a
8	res 0,b	res 0,c	res 0,d	res 0,e	res 0,h	res 0,l	res 0, (hl)	res 0,a	res 1,b	res 1,c	res 1,d	res 1,e	res 1,h	res 1,l	res 1, (hl)	res 1,a
9	res 2,b	res 2,c	res 2,d	res 2,e	res 2,h	res 2,l	res 2, (hl)	res 2,a	res 3,b	res 3,c	res 3,d	res 3,e	res 3,h	res 3,l	res 3, (hl)	res 3,a
A	res 4,b	res 4,c	res 4,d	res 4,e	res 4,h	res 4,l	res 4, (hl)	res 4,a	res 5,b	res 5,c	res 5,d	res 5,e	res 5,h	res 5,l	res 5, (hl)	res 5,a
B	res 6,b	res 6,c	res 6,d	res 6,e	res 6,h	res 6,l	res 6, (hl)	res 6,a	res 7,b	res 7,c	res 7,d	res 7,e	res 7,h	res 7,l	res 7, (hl)	res 7,a
C	set 0,b	set 0,c	set 0,d	set 0,e	set 0,h	set 0,l	set 0, (hl)	set 0,a	set 1,b	set 1,c	set 1,d	set 1,e	set 1,h	set 1,l	set 1, (hl)	set 1,a
D	set 2,b	set 2,c	set 2,d	set 2,e	set 2,h	set 2,l	set 2, (hl)	set 2,a	set 3,b	set 3,c	set 3,d	set 3,e	set 3,h	set 3,l	set 3, (hl)	set 3,a
E	set 4,b	set 4,c	set 4,d	set 4,e	set 4,h	set 4,l	set 4, (hl)	set 4,a	set 5,b	set 5,c	set 5,d	set 5,e	set 5,h	set 5,l	set 5, (hl)	set 5,a
F	set 6,b	set 6,c	set 6,d	set 6,e	set 6,h	set 6,l	set 6, (hl)	set 6,a	set 7,b	set 7,c	set 7,d	set 7,e	set 7,h	set 7,l	set 7, (hl)	set 7,a

IX Instructions (DD)

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0					inc b	dec b	ld b,n			add ix, bc			inc c	dec c	ld c,n	

1					inc d	dec d	ld d,n			add ix,de			inc e	dec e	ld e,n	
2		ld ix,n n	ld (n n),ix	inc ix	inc ixh	dec ixh	ld ixh, n			add ix, ix	ld ix, (nn)	dec ix	inc ixl	dec ixl	ld ixl, n	
3					inc (ix +d)	dec (ix +d)	ld (ix+ d),n			add ix, sp			inc a	dec a	ld a,n	
4	ld b,b	ld b,c	ld b,d	ld b,e	ld b,ix h	ld b,ix l	ld b,(i x+d)	ld b,a	ld c,b	ld c,c	ld c,d	ld c,e	ld c,ix h	ld c,ix l	ld c,(i x+d)	ld c,a
5	ld d,b	ld d,c	ld d,d	ld d,e	ld d,ix h	ld d,ix l	ld d,(i x+d)	ld d,a	ld e,b	ld e,c	ld e,d	ld e,e	ld e,ix h	ld e,ix l	ld e,(i x+d)	ld e,a
6	ld ixh, b	ld ixh, c	ld ixh, d	ld ixh, e	ld ixh, ixh	ld ixh, ixl	ld h,(i x+d)	ld ixh, a	ld ixl, b	ld ixl, c	ld ixl, d	ld ixl, e	ld ixl, ixh	ld ixl, ixl	ld l,(i x+d)	ld ixl, a
7	ld (ix+ d),b	ld (ix+ d),c	ld (ix+ d),d	ld (ix+ d),e	ld (ix+ d),h	ld (ix+ d),l		ld (ix+ d),a	ld a,b	ld a,c	ld a,d	ld a,e	ld a,ix h	ld a,ix l	ld a,(i x+d)	ld a,a
8	add a,b	add a,c	add a,d	add a,e	add a,i xh	add a,i xl	add a, (ix+d)	add a,a	adc a,b	adc a,c	adc a,d	adc a,e	adc a,i xh	adc a,i xl	adc a, (ix+d)	adc a,a
9	sub b	sub c	sub d	sub e	sub ixh	sub ixl	sub (ix +d)	sub a	sbc a,b	sbc a,c	sbc a,d	sbc a,e	sbc a,i xh	sbc a,i xl	sbc a, (ix+d)	sbc a,a
A	and b	and c	and d	and e	and ixh	and ixl	and (ix +d)	and a	xor b	xor c	xor d	xor e	xor ixh	xor ixl	xor (ix +d)	xor a
B	or b	or c	or d	or e	or ixh	or ixl	or (ix+ d)	or a	cp b	cp c	cp d	cp e	cp ixh	cp ixl	cp (ix+ d)	cp a
C												IX Bit				
E		pop ix		ex (s p),ix		push ix				jp (ix)						
F										ld sp,i x						

IX Bit Instructions (DDCB)

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	rlc (ix+d),b	rlc (ix+d),c	rlc (ix+d),d	rlc (ix+d),e	rlc (ix+d),h	rlc (ix+d),l	rlc (ix+d)	rlc (ix+d),a	rrc (ix+d),b	rrc (ix+d),c	rrc (ix+d),d	rrc (ix+d),e	rrc (ix+d),h	rrc (ix+d),l	rrc (ix+d)	rrc (ix+d),a
1	rl (ix+d),b	rl (ix+d),c	rl (ix+d),d	rl (ix+d),e	rl (ix+d),h	rl (ix+d),l	rl (ix+d)	rl (ix+d),a	rr (ix+d),b	rr (ix+d),c	rr (ix+d),d	rr (ix+d),e	rr (ix+d),h	rr (ix+d),l	rr (ix+d)	rr (ix+d),a
2	sla (ix+d),b	sla (ix+d),c	sla (ix+d),d	sla (ix+d),e	sla (ix+d),h	sla (ix+d),l	sla (ix+d)	sla (ix+d),a	sra (ix+d),b	sra (ix+d),c	sra (ix+d),d	sra (ix+d),e	sra (ix+d),h	sra (ix+d),l	sra (ix+d)	sra (ix+d),a
3	sll (ix+d),b	sll (ix+d),c	sll (ix+d),d	sll (ix+d),e	sll (ix+d),h	sll (ix+d),l	sll (ix+d)	sll (ix+d),a	srl (ix+d),b	srl (ix+d),c	srl (ix+d),d	srl (ix+d),e	srl (ix+d),h	srl (ix+d),l	srl (ix+d)	srl (ix+d),a
4	bit 0, (ix+d)	bit 0, (ix+d)	bit 0, (ix+d)	bit 0, (ix+d)	bit 0, (ix+d)	bit 0, (ix+d)	bit 0, (ix+d)	bit 0, (ix+d)	bit 1, (ix+d)	bit 1, (ix+d)	bit 1, (ix+d)	bit 1, (ix+d)	bit 1, (ix+d)	bit 1, (ix+d)	bit 1, (ix+d)	bit 1, (ix+d)
5	bit 2, (ix+d)	bit 2, (ix+d)	bit 2, (ix+d)	bit 2, (ix+d)	bit 2, (ix+d)	bit 2, (ix+d)	bit 2, (ix+d)	bit 2, (ix+d)	bit 3, (ix+d)	bit 3, (ix+d)	bit 3, (ix+d)	bit 3, (ix+d)	bit 3, (ix+d)	bit 3, (ix+d)	bit 3, (ix+d)	bit 3, (ix+d)
6	bit 4, (ix+d)	bit 4, (ix+d)	bit 4, (ix+d)	bit 4, (ix+d)	bit 4, (ix+d)	bit 4, (ix+d)	bit 4, (ix+d)	bit 4, (ix+d)	bit 5, (ix+d)	bit 5, (ix+d)	bit 5, (ix+d)	bit 5, (ix+d)	bit 5, (ix+d)	bit 5, (ix+d)	bit 5, (ix+d)	bit 5, (ix+d)
7	bit 6, (ix+d)	bit 6, (ix+d)	bit 6, (ix+d)	bit 6, (ix+d)	bit 6, (ix+d)	bit 6, (ix+d)	bit 6, (ix+d)	bit 6, (ix+d)	bit 7, (ix+d)	bit 7, (ix+d)	bit 7, (ix+d)	bit 7, (ix+d)	bit 7, (ix+d)	bit 7, (ix+d)	bit 7, (ix+d)	bit 7, (ix+d)
8	res 0, (ix+d), b	res 0, (ix+d), c	res 0, (ix+d), d	res 0, (ix+d), e	res 0, (ix+d), h	res 0, (ix+d), l	res 0, (ix+d)	res 0, (ix+d), a	res 1, (ix+d), b	res 1, (ix+d), c	res 1, (ix+d), d	res 1, (ix+d), e	res 1, (ix+d), h	res 1, (ix+d), l	res 1, (ix+d)	res 1, (ix+d), a
9	res 2, (ix+d), b	res 2, (ix+d), c	res 2, (ix+d), d	res 2, (ix+d), e	res 2, (ix+d), h	res 2, (ix+d), l	res 2, (ix+d)	res 2, (ix+d), a	res 3, (ix+d), b	res 3, (ix+d), c	res 3, (ix+d), d	res 3, (ix+d), e	res 3, (ix+d), h	res 3, (ix+d), l	res 3, (ix+d)	res 3, (ix+d), a
A	res 4, (ix+d), b	res 4, (ix+d), c	res 4, (ix+d), d	res 4, (ix+d), e	res 4, (ix+d), h	res 4, (ix+d), l	res 4, (ix+d)	res 4, (ix+d), a	res 5, (ix+d), b	res 5, (ix+d), c	res 5, (ix+d), d	res 5, (ix+d), e	res 5, (ix+d), h	res 5, (ix+d), l	res 5, (ix+d)	res 5, (ix+d), a
B	res 6, (ix+d), b	res 6, (ix+d), c	res 6, (ix+d), d	res 6, (ix+d), e	res 6, (ix+d), h	res 6, (ix+d), l	res 6, (ix+d)	res 6, (ix+d), a	res 7, (ix+d), b	res 7, (ix+d), c	res 7, (ix+d), d	res 7, (ix+d), e	res 7, (ix+d), h	res 7, (ix+d), l	res 7, (ix+d)	res 7, (ix+d), a
C	set 0, (ix+d), b	set 0, (ix+d), c	set 0, (ix+d), d	set 0, (ix+d), e	set 0, (ix+d), h	set 0, (ix+d), l	set 0, (ix+d)	set 0, (ix+d), a	set 1, (ix+d), b	set 1, (ix+d), c	set 1, (ix+d), d	set 1, (ix+d), e	set 1, (ix+d), h	set 1, (ix+d), l	set 1, (ix+d)	set 1, (ix+d), a
D	set 2, (ix+d), b	set 2, (ix+d), c	set 2, (ix+d), d	set 2, (ix+d), e	set 2, (ix+d), h	set 2, (ix+d), l	set 2, (ix+d)	set 2, (ix+d), a	set 3, (ix+d), b	set 3, (ix+d), c	set 3, (ix+d), d	set 3, (ix+d), e	set 3, (ix+d), h	set 3, (ix+d), l	set 3, (ix+d)	set 3, (ix+d), a
E	set 4, (ix+d), b	set 4, (ix+d), c	set 4, (ix+d), d	set 4, (ix+d), e	set 4, (ix+d), h	set 4, (ix+d), l	set 4, (ix+d)	set 4, (ix+d), a	set 5, (ix+d), b	set 5, (ix+d), c	set 5, (ix+d), d	set 5, (ix+d), e	set 5, (ix+d), h	set 5, (ix+d), l	set 5, (ix+d)	set 5, (ix+d), a

F	set 6, (ix+d), b	set 6, (ix+d), c	set 6, (ix+d), d	set 6, (ix+d), e	set 6, (ix+d), h	set 6, (ix+d), l	set 6, (ix+d)	set 6, (ix+d), a	set 7, (ix+d), b	set 7, (ix+d), c	set 7, (ix+d), d	set 7, (ix+d), e	set 7, (ix+d), h	set 7, (ix+d), l	set 7, (ix+d)	set 7, (ix+d), a
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IY Instructions (FD)

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0					inc b	dec b	ld b,n			add iy, bc			inc c	dec c	ld c,n	
1					inc d	dec d	ld d,n			add iy, de			inc e	dec e	ld e,n	
2		ld iy,n n	ld (n n),iy	inc iy	inc iyh	dec iyh	ld iyh, n			add iy, iy	ld iy, (nn)	dec iy	inc iyl	dec iyl	ld iyl, n	
3					inc (iy +d)	dec (iy +d)	ld (iy+ d),n			add iy, sp			inc a	dec a	ld a,n	
4	ld b,b	ld b,c	ld b,d	ld b,e	ld b,iy h	ld b,iy l	ld b,(i y+d)	ld b,a	ld c,b	ld c,c	ld c,d	ld c,e	ld c,iy h	ld c,iy l	ld c,(i y+d)	ld c,a
5	ld d,b	ld d,c	ld d,d	ld d,e	ld d,iy h	ld d,iy l	ld d,(i y+d)	ld d,a	ld e,b	ld e,c	ld e,d	ld e,e	ld e,iy h	ld e,iy l	ld e,(i y+d)	ld e,a
6	ld iyh, b	ld iyh, c	ld iyh, d	ld iyh, e	ld iyh, iyh	ld iyh, iyl	ld h,(i y+d)	ld iyh, a	ld iyl, b	ld iyl, c	ld iyl, d	ld iyl, e	ld iyl, iyh	ld iyl, iyl	ld l,(i y+d)	ld iyl, a
7	ld (iy+ d),b	ld (iy+ d),c	ld (iy+ d),d	ld (iy+ d),e	ld (iy+ d),h	ld (iy+ d),l		ld (iy+ d),a	ld a,b	ld a,c	ld a,d	ld a,e	ld a,iy h	ld a,iy l	ld a,(i y+d)	ld a,a
8	add a,b	add a,c	add a,d	add a,e	add a,i yh	add a,i yl	add a, (iy+d)	add a,a	adc a,b	adc a,c	adc a,d	adc a,e	adc a,i yh	adc a,i yl	adc a, (iy+d)	adc a,a
9	sub b	sub c	sub d	sub e	sub iyh	sub iyl	sub (iy +d)	sub a	sbc a,b	sbc a,c	sbc a,d	sbc a,e	sbc a,i yh	sbc a,i yl	sbc a, (iy+d)	sbc a,a
A	and b	and c	and d	and e	and iyh	and iyl	and (iy +d)	and a	xor b	xor c	xor d	xor e	xor iyh	xor iyl	xor (iy +d)	xor a
B	or b	or c	or d	or e	or iyh	or iyl	or (iy+ d)	or a	cp b	cp c	cp d	cp e	cp iyh	cp iyl	cp (iy+ d)	cp a

C												IY Bit				
E		pop iy		ex (s p),iy		push iy				jp (iy)						
F										ld sp,i y						

IY Bit Instructions (FDCB)

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	rlc (iy+d),b	rlc (iy+d),c	rlc (iy+d),d	rlc (iy+d),e	rlc (iy+d),h	rlc (iy+d),l	rlc (iy+d)	rlc (iy+d),a	rrc (iy+d),b	rrc (iy+d),c	rrc (iy+d),d	rrc (iy+d),e	rrc (iy+d),h	rrc (iy+d),l	rrc (iy+d)	rrc (iy+d),a
1	rl (iy+d),b	rl (iy+d),c	rl (iy+d),d	rl (iy+d),e	rl (iy+d),h	rl (iy+d),l	rl (iy+d)	rl (iy+d),a	rr (iy+d),b	rr (iy+d),c	rr (iy+d),d	rr (iy+d),e	rr (iy+d),h	rr (iy+d),l	rr (iy+d)	rr (iy+d),a
2	sla (iy+d),b	sla (iy+d),c	sla (iy+d),d	sla (iy+d),e	sla (iy+d),h	sla (iy+d),l	sla (iy+d)	sla (iy+d),a	sra (iy+d),b	sra (iy+d),c	sra (iy+d),d	sra (iy+d),e	sra (iy+d),h	sra (iy+d),l	sra (iy+d)	sra (iy+d),a
3	sll (iy+d),b	sll (iy+d),c	sll (iy+d),d	sll (iy+d),e	sll (iy+d),h	sll (iy+d),l	sll (iy+d)	sll (iy+d),a	srl (iy+d),b	srl (iy+d),c	srl (iy+d),d	srl (iy+d),e	srl (iy+d),h	srl (iy+d),l	srl (iy+d)	srl (iy+d),a
4	bit 0, (iy+d)	bit 0, (iy+d)	bit 0, (iy+d)	bit 0, (iy+d)	bit 0, (iy+d)	bit 0, (iy+d)	bit 0, (iy+d)	bit 0, (iy+d)	bit 1, (iy+d)	bit 1, (iy+d)	bit 1, (iy+d)	bit 1, (iy+d)	bit 1, (iy+d)	bit 1, (iy+d)	bit 1, (iy+d)	bit 1, (iy+d)
5	bit 2, (iy+d)	bit 2, (iy+d)	bit 2, (iy+d)	bit 2, (iy+d)	bit 2, (iy+d)	bit 2, (iy+d)	bit 2, (iy+d)	bit 2, (iy+d)	bit 3, (iy+d)	bit 3, (iy+d)	bit 3, (iy+d)	bit 3, (iy+d)	bit 3, (iy+d)	bit 3, (iy+d)	bit 3, (iy+d)	bit 3, (iy+d)
6	bit 4, (iy+d)	bit 4, (iy+d)	bit 4, (iy+d)	bit 4, (iy+d)	bit 4, (iy+d)	bit 4, (iy+d)	bit 4, (iy+d)	bit 4, (iy+d)	bit 5, (iy+d)	bit 5, (iy+d)	bit 5, (iy+d)	bit 5, (iy+d)	bit 5, (iy+d)	bit 5, (iy+d)	bit 5, (iy+d)	bit 5, (iy+d)
7	bit 6, (iy+d)	bit 6, (iy+d)	bit 6, (iy+d)	bit 6, (iy+d)	bit 6, (iy+d)	bit 6, (iy+d)	bit 6, (iy+d)	bit 6, (iy+d)	bit 7, (iy+d)	bit 7, (iy+d)	bit 7, (iy+d)	bit 7, (iy+d)	bit 7, (iy+d)	bit 7, (iy+d)	bit 7, (iy+d)	bit 7, (iy+d)
8	res 0, (iy+d), b	res 0, (iy+d), c	res 0, (iy+d), d	res 0, (iy+d), e	res 0, (iy+d), h	res 0, (iy+d), l	res 0, (iy+d)	res 0, (iy+d), a	res 1, (iy+d), b	res 1, (iy+d), c	res 1, (iy+d), d	res 1, (iy+d), e	res 1, (iy+d), h	res 1, (iy+d), l	res 1, (iy+d)	res 1, (iy+d), a
9	res 2, (iy+d), b	res 2, (iy+d), c	res 2, (iy+d), d	res 2, (iy+d), e	res 2, (iy+d), h	res 2, (iy+d), l	res 2, (iy+d)	res 2, (iy+d), a	res 3, (iy+d), b	res 3, (iy+d), c	res 3, (iy+d), d	res 3, (iy+d), e	res 3, (iy+d), h	res 3, (iy+d), l	res 3, (iy+d)	res 3, (iy+d), a

A	res 4, (iy+d), b	res 4, (iy+d), c	res 4, (iy+d), d	res 4, (iy+d), e	res 4, (iy+d), h	res 4, (iy+d), l	res 4, (iy+d)	res 4, (iy+d), a	res 5, (iy+d), b	res 5, (iy+d), c	res 5, (iy+d), d	res 5, (iy+d), e	res 5, (iy+d), h	res 5, (iy+d), l	res 5, (iy+d)	res 5, (iy+d), a
B	res 6, (iy+d), b	res 6, (iy+d), c	res 6, (iy+d), d	res 6, (iy+d), e	res 6, (iy+d), h	res 6, (iy+d), l	res 6, (iy+d)	res 6, (iy+d), a	res 7, (iy+d), b	res 7, (iy+d), c	res 7, (iy+d), d	res 7, (iy+d), e	res 7, (iy+d), h	res 7, (iy+d), l	res 7, (iy+d)	res 7, (iy+d), a
C	set 0, (iy+d), b	set 0, (iy+d), c	set 0, (iy+d), d	set 0, (iy+d), e	set 0, (iy+d), h	set 0, (iy+d), l	set 0, (iy+d)	set 0, (iy+d), a	set 1, (iy+d), b	set 1, (iy+d), c	set 1, (iy+d), d	set 1, (iy+d), e	set 1, (iy+d), h	set 1, (iy+d), l	set 1, (iy+d)	set 1, (iy+d), a
D	set 2, (iy+d), b	set 2, (iy+d), c	set 2, (iy+d), d	set 2, (iy+d), e	set 2, (iy+d), h	set 2, (iy+d), l	set 2, (iy+d)	set 2, (iy+d), a	set 3, (iy+d), b	set 3, (iy+d), c	set 3, (iy+d), d	set 3, (iy+d), e	set 3, (iy+d), h	set 3, (iy+d), l	set 3, (iy+d)	set 3, (iy+d), a
E	set 4, (iy+d), b	set 4, (iy+d), c	set 4, (iy+d), d	set 4, (iy+d), e	set 4, (iy+d), h	set 4, (iy+d), l	set 4, (iy+d)	set 4, (iy+d), a	set 5, (iy+d), b	set 5, (iy+d), c	set 5, (iy+d), d	set 5, (iy+d), e	set 5, (iy+d), h	set 5, (iy+d), l	set 5, (iy+d)	set 5, (iy+d), a
F	set 6, (iy+d), b	set 6, (iy+d), c	set 6, (iy+d), d	set 6, (iy+d), e	set 6, (iy+d), h	set 6, (iy+d), l	set 6, (iy+d)	set 6, (iy+d), a	set 7, (iy+d), b	set 7, (iy+d), c	set 7, (iy+d), d	set 7, (iy+d), e	set 7, (iy+d), h	set 7, (iy+d), l	set 7, (iy+d)	set 7, (iy+d), a